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INFORMATION PROCESSING APPARATUS WITH AUTOMATIC DETECTION AND INTERRUPTION OF DEFECTIVE IMAGE FORMING OPERATIONS

Abstract

An information processing apparatus, which communicates with an image forming apparatus and a reader, including: a display; and a controller configured to: obtain read image data output from the reader; determine an error of the read image data based on the read image data and reference data; cancel image forming operation of the image forming apparatus in a case where the error is determined in succession a predetermined number of times; cancel reading operation of the reader in the case where the error is determined in succession the predetermined number of times; notify, on the display, the cancellation of the image forming operation and the reading operation in the case where the error is determined in succession the predetermined number of times; and receive a user's instruction information about the predetermined number of times.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/607,062 filed Mar. 15, 2024, which is a continuation of U.S. application Ser. No. 18/349,881 filed Jul. 10, 2023, now issued as U.S. Pat. No. 11,972,314 on Apr. 30, 2024, which is a continuation of application Ser. No. 17/176,985 filed Feb. 16, 2021, now issued as U.S. Pat. No. 11,727,235 on Aug. 15, 2023; and claims priority under 35 U.S.C. § 119 to Japan Application JP 2020-028538 filed in Japan Feb. 21, 2020; and the contents of all of which are incorporated herein by reference as if set forth in full.

BACKGROUND OF THE INVENTION
Field of the Invention
[0002] The present invention relates to an inspection technology for inspecting an output image formed by an image forming apparatus on a recording medium.
Description of the Related Art
[0003] Inspection work on a printed material has been performed by a worker's visual inspection based on experience of the worker. A visual check of an enormous volume of printed materials, however, is a heavy burden on the worker and also invites a high probability of mistakes in checking. Further, a visual check of printed materials output at high speed from an image forming apparatus cannot be said to be high in work efficiency. Automatic inspection in synchronization with printing of a printed material output from a high-speed image forming apparatus compatible with various media has therefore been sought after.
[0004] In recent years, there has been developed a system in which a printed material output from an image forming apparatus is read by a sensor, image data obtained by the reading is subjected to image processing, and the processed image data is compared to print data that is original data, to thereby detect defects caused at a time of printing, such as stains, blank spots, and skew feeding in printing. The system further controls a finisher provided with a plurality of discharge portions, based on a determination result of a printed material, to change a discharge position for a printed product that is a normally printed material and for a printed product that is a printed material having a detected defect. With this technique, a printed product that is a printed material with a printing defect and a printed product that is a printed material without a printing defect can be sorted (Japanese Patent Application Laid-Open No. 2010-42521).
[0005] In the inspection system of the related art, when specks adhere to the reading sensor or a similar trouble occurs, shadows of the specks or the like appear on image data generated by reading a printed material. Data inconsistency due to an effect of the specks or the like is consequently inevitable in a comparison between the image data read in an inspection process and print data that is original data, and an inspection result indicating an error is returned. Then, the inspection process

cannot be resumed unless the specks that are the cause of the error are removed, or, in the case of a scratch or the like in a reading portion, the scratch is removed by a repair, and production ceases for that duration.
[0006] Dust accidentally mixed in at the time of reading a printed material, instead of the adhering dust described above, produce an inspection result indicating an error as well when data inconsistency is caused by the mixed-in dust. The accidentally mixed-in dust may not affect a subsequent inspection process. However, the inspection process is stopped because whether the dust are fixed dust or temporarily present dust cannot be determined, and production ceases for that duration.
[0007] Precision at which inspection is to be performed and items to be inspected vary depending on a user's request and contents of data to be inspected. In inspection of a printed material in which various data contents are mixed, some inspection items may cause a significant drop in processing speed. There is a technology that can improve a degree of certainty and productivity of the inspection process by solving the above-mentioned problem (Japanese Patent Application Laid-Open No. 2013-252632).
[0008] In an information processing apparatus of the related art, image forming operation continues until all images corresponding to a print job finish being formed, despite an inspection result indicating an error. That is, the information processing apparatus of the related art cannot stop the image forming operation by an image forming apparatus even when successive inspection results indicate errors. The information processing apparatus of the related art therefore has a possibility that printed products whose inspection results indicate errors are kept output when, for example, there is an abnormality in the image forming apparatus or a reader. That may disadvantage a user by wasting a recording medium, which is assets of the user.
SUMMARY OF THE INVENTION
[0009] It is therefore an object of the present invention to inhibit continuation of output of printed products that are determined to be errors as a result of inspection.
[0010] There is provided an information processing apparatus which is communicable in communication to and from an image forming apparatus configured to form an image on a sheet, and in communication to and from a reader configured to read the image on the sheet, the reader being connected to the image forming apparatus, the information processing apparatus comprising: a display configured to display an inspection result; and a controller configured to: obtain read image data output from the reader; determine an error of the read image data based on the read image data and reference data registered in advance; cancel image forming operation of forming an image by the image forming apparatus in a case in which the error is determined in succession a predetermined number of times; cancel reading operation of reading an image by the reader in the case in which the error is determined in succession the predetermined number of times; notify, on the display, the cancellation of the image forming operation and the reading operation in the case in which the error is determined in succession the predetermined number of times; and receive a user's instruction information about the predetermined number of times.
[0011] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0012] FIG. **1** is an overall view of a hardware configuration of a printing system.
[0013] FIG. **2** is a block diagram of a system configuration of the printing system.
[0014] FIG. **3**A and FIG. **3**B are each a diagram for illustrating an example of processing of an inspection image that is performed with respect to a vertical line.

[0015] FIG. **4**A and FIG. **4**B are each a group of graphs for showing an example of processing of profile calculation that is performed with respect to a vertical line.
[0016] FIG. **5** is a cross-sectional view of an image forming apparatus.
[0017] FIG. **6** is a flow chart of an inspection process.
[0018] FIG. **7** is a flow chart of a factor identification process.
[0019] FIG. **8** is a flow chart of a factor identification process in a second embodiment.
[0020] FIG. **9** is a flow chart of a factor identification process in a third embodiment.
[0021] FIG. **10** is a flow chart of a factor identification process in a fourth embodiment.
[0022] FIG. **11** is a flow chart of improvement operation in a fifth embodiment.
[0023] FIG. **12** is a schematic diagram of a setting screen.
[0024] FIG. **13** is a schematic diagram of a cancellation screen.

DESCRIPTION OF THE EMBODIMENTS

First Embodiment

[0025] Embodiments are described below with reference to the accompanying drawings. In the following description, an external controller **102** is referred to as “image processing controller,” “digital front end,” “print server,” or “DFE” in some places. An image forming apparatus **101** is referred to as “multi-function printer,” “multi-function peripheral,” or “MFP” in some places.

(Printing System)

[0026] FIG. **1** is an overall view of a hardware configuration of a printing system **100**. The printing system **100** includes the image forming apparatus **101** and the external controller **102**. The image forming apparatus **101** and the external controller **102** are connected in a manner that allows communication via an internal LAN **105** and a video cable **106**. The external controller **102** is connected to a client personal computer (hereinafter referred to as “PC”) **103** in a manner that allows communication via an external LAN **104**. The PC **103** issues a print instruction to the external controller **102**.

[0027] A printer driver having a function of converting print data that is original data into a print description language processable by the external controller **102** is installed in the PC **103**. A user can issue a print instruction to the external controller **102** via the printer driver from various applications. The printer driver transmits print data to the external controller **102** based on the print instruction from the user. When receiving the print instruction from the PC **103**, the external controller **102** performs data analysis and rasterization processing, and executes transmission of the print data and issuing of a print instruction to the image forming apparatus **101**.

[0028] The image forming apparatus **101** is described next. A plurality of devices having different functions are connected to the image forming apparatus **101** so that complicate printing processing, for example, binding, is executable. The image forming apparatus **101** includes a printer **107**, an inserter **108**, an inspection device **109**, a high capacity stacker **110**, and a finisher **111**.

[0029] The printer **107** as an image forming unit is configured to form an image, with the use of a toner, on paper or another recording medium (hereinafter referred to as “sheet”) conveyed from a feed portion, which is located in a lower part of the printer **107**. A configuration and operation principle of the printer **107** are as follows. A surface of a photosensitive drum is uniformly charged to a predetermined electric potential by a charging unit. A beam of light (hereinafter referred to as “laser light”) modulated based on print data (image data that is original data) is reflected by a rotary polygon mirror, and the reflected light irradiates the surface of the photosensitive drum as scanning light. The uniformly charged surface of the photosensitive drum is exposed to the laser light, with the result that an electrostatic latent image is formed on the surface of the photosensitive drum. The electrostatic latent image formed on the surface of the photosensitive drum is developed with a toner to form a toner image. The toner image formed on the surface of the photosensitive drum is transferred onto the sheet by a transfer drum. This series of steps of an image forming process is sequentially executed with the use of a yellow (Y) toner, a magenta (M) toner, a cyan (C) toner, and a black (K) toner, to thereby form a full-color image on the sheet. The sheet with the full-color image formed thereon is conveyed to a first fixing unit **311** (FIG. **5**). The first fixing unit **311** is configured to apply heat and pressure to the sheet to which the toner image has been transferred, to thereby fix the toner image to the sheet.

[0030] The inserter **108** is configured to insert an insertion sheet in a suitable place in relation to a group of sheets printed by and conveyed from the printer **107**. The inspection device **109** reads an image of a sheet conveyed thereto, and determines whether the printed image is normal by a comparison of image data obtained by the reading to correct image data (reference data) stored in advance in a memory. The correct image data is registered by any method. The correct image data may be generated by, for example, performing RIP processing on print data forwarded from the PC **103**, or reading, with the inspection device **109**, a printed product on which printing has been performed in advance. A large number of sheets can be placed on the high capacity stacker **110**. The finisher **111** is configured to perform post-processing on a sheet conveyed thereto. The post-processing is, for example, stapling, punching, saddle stitch binding, and the like. A sheet with an image formed thereon is discharged to a discharge tray of the finisher **111**.

[0031] The printing system **100** illustrated in FIG. **1** has a configuration in which the external controller **102** is connected to the image forming apparatus **101**. The printing system **100**, however, is not limited to the configuration in which the external controller **102** is connected to the image forming apparatus **101**. For example, the printing system **100** may be configured so that the image forming apparatus **101** is connected to the external LAN **104** to receive print data processable by the image forming apparatus **101** from the PC **103** via the external LAN **104**. In this case, data analysis and rasterization processing are performed in the image forming apparatus **101**, and the image forming apparatus **101** executes printing processing.

[0032] FIG. **2** is a block diagram of a system configuration of the printing system **100**. A configuration of the printer **107** of the image forming apparatus **101** is described first. The printer **107** includes a communication I/F (COMM I/F) **217**, a LAN I/F **218**, a video I/F **220**, an HDD **221**, a CPU **222**, a memory **223**, an operation portion **224**, and a display portion (display) **225**. The printer **107** further includes an original exposure portion **226**, a laser exposure portion **227**, an image forming portion **228**, a fixing portion **229**, and a feed portion **230**. The components are connected to one another via a system bus **231**.

[0033] The communication I/F **217** is connected to, via a communication cable **254**, the inserter **108**, the inspection device **109**, the high capacity stacker **110**, and the finisher **111**. Communication for controlling the devices is held via the communication cable **254**. The LAN I/F **218** is connected to the external controller **102** via the internal LAN **105**.

[0034] Print data is communicated between the external controller **102** and the printer **107** via the internal LAN **105**. The video I/F **220** is connected to the external controller **102** via the video cable **106**. Image data is communicated between the external controller **102** and the printer **107** via the video cable **106**. The HDD **221** is a storage device in which programs and data are stored. The CPU **222** is configured to perform image processing control and printing control in a comprehensive manner, based on a program stored in the HDD **221**. The memory **223** stores programs and image data required when the CPU **222** executes various types of processing, and operates as a work area. The operation portion **224** is configured to receive input of various settings and an instruction on operation from the user. The display portion **225** is configured to display settings information of the printer **107** and a processing status of a print job.

[0035] The original exposure portion **226** executes processing of reading an original when a copy function or a scan function is used. The user places an original on platen glass (an original table), and an image of the original irradiated with light from an exposure lamp is taken with a CCD camera, to thereby read image data of the original.

[0036] The laser exposure portion **227** is configured to perform primary charging in which the surface of the photosensitive drum is uniformly charged, and laser exposure. The laser exposure portion **227** first charges the surface of the photosensitive drum to a uniform negative electric potential (primary charging). The laser exposure portion **227** next uses a laser driver to emit laser light in a pattern based on image data, and irradiates the photosensitive drum with the laser light at a reflection angle adjusted by the polygon mirror. Negative electric charges in a part irradiated with the laser light are thus neutralized and an electrostatic latent image is consequently formed on the surface of the photosensitive drum.

[0037] The image forming portion **228** includes a developing unit, a transfer unit, and a toner replenishment portion. The developing unit is configured to cause a negatively charged toner from a developing cylinder to adhere to the electrostatic latent image on the surface of the photosensitive drum, to thereby turn the electrostatic latent image into a toner image. The transfer unit includes a primary transfer roller and a secondary transfer outer roller. The transfer unit is configured to apply a positive electric potential to the primary transfer roller to transfer the toner image on the surface of the photosensitive drum to a transfer belt (primary transfer). The transfer unit is also configured to apply a positive electric potential to the secondary transfer outer roller to transfer the toner image on the transfer belt to a sheet (secondary transfer).

[0038] The fixing portion **229** includes a heater, a fixing belt, and a pressure belt. The fixing portion **229** is configured to apply pressure and heat to the sheet at a nip between the fixing belt and the pressure belt to melt the toner image and thus fix the toner image to the sheet. The feed portion **230** is configured to control sheet feeding operation and sheet conveying operation with rollers and various sensors.

[0039] A configuration of the inserter **108** of the image forming apparatus **101** is described next. The inserter **108** includes a communication I/F **232**, a CPU **233**, a memory **234**, and a feed controller **235**. The components are connected to one another via a system bus **236**. The communication I/F **232** is connected to the printer **107** via the communication cable **254**. The inserter **108** holds communication required for control to and from the printer **107** via the communication cable **254**. The CPU **233** is configured to perform various types of control required for feeding in accordance with a control program stored in the memory **234**. The memory **234** is a storage device in which the control program is stored. The feed controller **235** is configured to control a feed portion, rollers, and sensors of the inserter **108** following an instruction from the CPU **233**, to control the feeding of a sheet conveyed from the printer **107**.

[0040] A configuration of the inspection device **109** of the image forming apparatus **101** is described next. The inspection device **109** includes a communication I/F **237**, a CPU **238**, a memory **239**, an imaging portion **240**, a display portion **241**, and an operation portion **242**. The components are connected to one another via a system bus **243**. The communication I/F **237** is connected to the printer **107** via the communication cable **254**. The inspection device **109** holds communication required for control to and from the printer **107** via the communication cable **254**. The CPU **238** is configured to perform various types of control required for inspection operation in accordance with a control program stored in the memory **239**. The memory **239** is a storage device in which the control program is stored. The imaging portion **240** includes image reading sensors **331** and **332** (FIG. **5**) as a reading unit configured to read an image that is formed on a sheet. The imaging portion **240** is configured to take an image of a conveyed sheet following an instruction of the CPU **238**. The CPU **238** compares read image data obtained through the image taking by the imaging portion **240** to correct image data stored in the memory **239** to determine whether the image formed on the sheet is normal. The display portion **241** is configured to display an inspection result and a setting screen **141** (FIG. **12**). The operation portion **242** is operated by the user to receive a change to settings of the inspection device **109** and an instruction to register correct image data.

[0041] The CPU **238** as a determination unit determines whether there is an image defect in the image formed on the sheet, based on reading results (detection results) of the image reading sensors **331** and **332** (FIG. **5**). An inspection process in which an image is inspected for an image defect is described below with reference to FIG. **3**A, FIG. **3**B, FIG. **4**A, and FIG. **4**B. FIG. **3**A and FIG. **3**B are each a diagram for illustrating an example of processing that is executed when a vertical line appears in an image to be inspected (inspection image). FIG. **3**A is a diagram for illustrating a sheet S on which a black vertical line is formed and an actual measurement density profile of the sheet S. FIG. **3**B is a diagram for illustrating a sheet S on which a white vertical line is formed and an actual measurement density profile of the sheet S. There are various methods of inspection of an output image. The method of inspection of the first embodiment is not limited to the following method. Various inspection methods may be employed in the first embodiment. Here, examples of vertical line detection are described. Detection of a vertical line in an output image is executed by the CPU **238** through a comparison of an actual measurement density profile. The "actual measurement density profile" here means what is obtained by measuring an image density at each position in a width direction of the sheet S being conveyed, and integrating image densities in a sheet conveyance direction at the position.

[0042] The CPU **238** first generates the actual measurement density profile by integration of an image on the sheet S that has been read by the image reading sensor **331**, along the sheet conveyance direction as illustrated in FIG. **3**A and FIG. **3**B. The CPU **238** compares the generated actual measurement density profile to a profile of image data that is original data, to thereby inspect the image on the sheet S. Two streak types are assumed here, with one being a black vertical line, which is a longitudinal line running in a white part as illustrated in FIG. **3**A, and the other being a white vertical line, which is a blank streak caused by a missing part of the image as illustrated in FIG. **3**B.

[0043] FIG. **4**A and FIG. **4**B are each a group of graphs for showing an example of processing of profile calculation that is performed with respect to a vertical line. FIG. **4**A is a group of graphs for showing a predicted density profile, an actual measurement density profile, and an absolute value of a difference for the black vertical line. FIG. **4**B is a group of graphs for showing a predicted density profile, an actual measurement density profile, and an absolute value of a difference for the white vertical line. The CPU **222** of the printer **107** transmits image data that is original data to the CPU **238** of the inspection device **109**. The CPU **238** of the inspection device **109** generates, in advance, a predicted density profile by integration in a vertical direction based on the received image data that is original data. The CPU **238** calculates a difference between the generated predicted density profile and the actual measurement density profile. The CPU **238** determines whether the sheet S on which the image has been formed by the printer **107** is a defective sheet based on whether the absolute value of the difference between the predicted density profile and the actual measurement density profile exceeds a predetermined threshold value. Sheet information and the threshold value (inspection threshold value) can be set in a discretionary manner.

[0044] A configuration of the high capacity stacker **110** of the image forming apparatus **101** is described next. The high capacity stacker **110** includes a communication I/F **244**, a CPU **245**, a memory **246**, and a discharge controller **247**. The components are connected to one another via a system bus **248**. The communication I/F **244** is connected to the printer **107** via the communication cable **254**. The high capacity stacker **110** holds communication required for control to and from the printer **107** via the communication cable **254**. The CPU **245** is configured to perform various types of control required to discharge a sheet in accordance with a control program stored in the memory **246**. The memory **246** is a storage device in which the control program is stored. The discharge controller **247** is configured to perform control for conveying a conveyed sheet to a sheet stacking tray **341** (FIG. **5**), an escape tray (purge tray) **346** (FIG. **5**), or the finisher **111**, following an instruction from the CPU **245**.

[0045] A configuration of the finisher **111** of the image forming apparatus **101** is described next. The finisher **111** includes a communication I/F **249**, a CPU **250**, a memory **251**, a discharge controller **252**, and a finishing process portion **253**. The components are connected to one another via a system bus **255**. The communication I/F **249** is connected to the printer **107** via the communication cable **254**. The finisher **111** holds communication required for control to and from the printer **107** via the communication cable **254**. The CPU **250** is configured to perform various types of control required for finishing and discharge in accordance with a control program stored in the memory **251**. The memory **251** is a storage device in which the control program is stored. The discharge controller **252** is configured to control the conveyance and discharge of a sheet following an instruction from the CPU **250**. The finishing process portion **253** is configured to control finishing processing, which is stapling, punching, saddle stitch binding, or the like, following an instruction from the CPU **250**.

[0046] A configuration of the external controller **102** is described next. The external controller **102** includes a CPU **208**, a memory **209**, an HDD **210**, a keyboard **211**, a display **212**, a LAN I/F **213**. a LAN I/F **214**, and a video I/F **215**. The components are connected to one another via a system bus **216**. The CPU **208** is configured to execute, in a comprehensive manner, processing including reception of print data from the PC **103**, RIP processing, and transmission of print data that is original data to the printer **107**, based on programs and data stored in the HDD **210**. The memory **209** stores programs and data required when the CPU **208** executes various types of processing, and operates as a work area. The HDD **210** stores programs and data required for operation of printing processing and the like.

[0047] The keyboard **211** is used by the user in order to input an operation instruction to the external controller **102**. The display **212** is configured to display information of an application that is run in the external controller **102** and other information with a video signal of a still image or a moving image. The LAN I/F **213** is connected to the PC **103** via the external LAN **104**. The external controller **102** communicates a print instruction and the like to and from the PC **103** via the external LAN **104**. The LAN I/F **214** is connected to the printer **107** via the internal LAN **105**. The external controller **102** communicates a print instruction and the like to and from the printer **107** via the internal LAN **105**. The video I/F **215** is connected to the printer **107** via the video cable **106**. The external controller **102** communicates print data that is original data and the like to and from the printer **107** via the video cable **106**.

[0048] A configuration of the PC **103** is described next. The PC **103** includes a CPU **201**, a memory **202**, an HDD **203**, a keyboard **204**, a display **205**, and a LAN I/F **206**. The components are connected to one another via a system bus **207**. The CPU **201** is configured to generate print data and execute a print instruction based on a document processing program or the like stored in the HDD **203**. The CPU **201** is also configured to control the devices connected to the system bus **207** in a comprehensive manner. The memory **202** stores programs and data required when the CPU **201** executes various types of processing, and operates as a work area. The HDD **203** stores programs and data required for operation of printing processing and the like. The keyboard **204** is used by the user in order to input an operation instruction to the PC **103**. The display **205** is configured to display information of an application that is run on the PC **103** and other information with a video signal of a still image or a moving image. The LAN I/F **206** is connected to the external LAN **104**. The PC **103** communicates a print instruction and the like to and from the external controller **102** via the external LAN **104**.

[0049] The external controller **102** and the printer **107** are connected by the internal LAN **105** and the video cable **106**. Connection between the external controller **102** and the printer **107**, however, may be configured any way as long as data required for printing can be transmitted and received. For example, the external controller **102** and the printer **107** may be connected by the video cable **106** alone. Each of the memory **202**, the memory **209**, the memory **223**, the memory **234**, the memory **239**, the memory **246**, and the memory **251** can be any storage device for storing data and a program. Those memories may be, for example, volatile RAMs, non-volatile ROMs, built-in HDDs, external HDDs, or USB memories.

[0050] FIG. **5** is a cross-sectional view of the image forming apparatus **101**. The printer **107** is configured to form an image on a sheet in accordance with image data that is original data. The printer **107** is provided with feed decks **301** and **302**. The feed decks **301** and **302** each can accommodate various types of sheets. The feed decks **301** and **302** are each configured to separate the topmost sheet among accommodated sheets one by one, and conveys the separated sheet to a sheet conveying path **303**. Developing stations **304**, **305**, **306**, and **307** are configured to form toner images on surfaces of photosensitive drums with the use of a yellow (Y) toner, a magenta (M) toner, a cyan (C) toner, and a black (K) toner, respectively, in order to form a color image. The toner images formed on the surfaces of the photosensitive drums are transferred by primary transfer onto an intermediate transfer belt **308**. The intermediate transfer belt **308** is configured to rotate clockwise in FIG. **5**, and the toner images are transferred at a secondary transfer position **309** onto the sheet conveyed from the sheet conveying path **303**. The display portion **225** is configured to display a printing status of the image forming apparatus **101** and information for setting the image forming apparatus **101**.

[0051] The first fixing unit **311** is configured to fix the toner images to the sheet. The first fixing unit **311** includes a pressure roller and a heating roller. The toners are melted under pressure by the passing of the sheet through a nip between the pressure roller and the heating roller, and the toner images are thus fixed to the sheet. The sheet having passed through the first fixing unit **311** travels on a sheet conveying path **312** to be conveyed to a sheet conveying path **315**. Some types of sheets require further heating and pressuring to fix the toner images to the sheet. In that case, the sheet having passed through the first fixing unit **311** is conveyed to a second fixing unit **313** through a sheet conveying path on an upper side. The second fixing unit **313** applies further heat and pressure to the sheet, to thereby fix the toner images to the sheet. The sheet having passed through the second fixing unit **313** travels on a sheet conveying path **314** to be conveyed to the sheet conveying path **315**. When an image forming mode is double-sided printing, the sheet is conveyed to a sheet reversing path **316**. The sheet is reversed in terms of front and back by the sheet reversing path **316**, and is then conveyed to a double-sided printing conveying path **317**. The sheet is conveyed from the double-sided printing conveying path **317** to the sheet conveying path **303**, and toner images are transferred onto the back side (second side) of the sheet at the secondary transfer position **309**.

[0052] The inserter **108** is configured to insert an insertion sheet. The inserter **108** includes an inserter tray **321** on which the insertion sheet is placed. The inserter **108** conveys the insertion sheet from the inserter tray **321** to a sheet conveying path **333** through a sheet conveying path **322**. The insertion sheet can thus be inserted in a suitable place in relation to a series of sheets conveyed from the printer **107**, for example, before, among, or after the series of sheets. The inserter **108** includes a conveyance roller provided with a side-shift roller and a skew roller. The inserter **108** has a shift mechanism, which, when a sheet deviates from a reference position in a direction (side direction) orthogonal to the sheet conveyance direction, uses the side-shift roller and the skew roller to correct the deviation of the sheet and bring the sheet back to the reference position.

[0053] The sheet having passed through the inserter **108** is conveyed to the inspection device **109**. The image reading sensors **331** and **332** are arranged so as to face each other in the inspection device **109**. The image reading sensor **331** is configured to read a top side (one side) of the sheet being conveyed on the sheet conveying path **333**. The image reading sensor **332** is configured to read a bottom side (the other side) of the sheet being conveyed on the sheet conveying path **333**. The inspection device **109** uses the image reading sensors **331** and **332** to read images of the sheet at the time when the sheet conveyed to the sheet conveying path **333** reaches a predetermined position. The inspection device **109** determines whether the images formed by the printer **107** are normal based on reading results (detection results) of the image reading sensors **331** and **332**. An inspection result by the inspection device **109** is displayed on the display portion **241**.

[0054] A large number of sheets can be placed in the high capacity stacker **110**. The high capacity stacker **110** includes the sheet stacking tray **341** as a tray on which sheets are placed. The sheet having passed through the inspection device **109** travels on a sheet conveying path **344** to be placed in the high capacity stacker **110**. The sheet is placed on the sheet stacking tray **341** from the sheet conveying path **344** via a sheet conveying path **345**. The high capacity stacker **110** further includes the escape tray **346** as a discharge tray. A sheet determined by the inspection device **109** to be defective is discharged to the escape tray **346** from the sheet conveying path **344** via a sheet conveying path **347**.

[0055] A sheet to be conveyed to the finisher **111** as a post-processing device connected downstream of the high capacity stacker **110** in the sheet conveyance direction is conveyed to the finisher **111** from the sheet conveying path **344** via a sheet conveying path **348**. The high capacity stacker **110** is provided with a reverse portion **349** configured to reverse a sheet. The reverse portion **349** is used to place a sheet on the sheet stacking tray **341**. When a sheet is to be placed on the sheet stacking tray **341** so that a direction in which the sheet is input and a direction in which the sheet is output are the same, the sheet is reversed once by the reverse portion **349**. The operation of reversing a sheet by the reverse portion **349** is not executed for a sheet to be conveyed to the escape tray **346** or the finisher **111** because the sheet is discharged as it is without being reversed prior to being placed on the tray.

[0056] The finisher **111** is configured to perform, on a sheet conveyed thereto, post-processing (hereinafter referred to as "finishing processing") that corresponds to a function specified by the user. Specifically, the finisher **111** has a finishing function including stapling (one-place stapling, two-place stapling), punching (two holes, three holes), saddle stitch binding, and others. The finisher **111** has two discharge trays **351** and **352**. The sheet is discharged to the discharge tray **351** via a sheet conveying path **353**. However, stapling and other types of finishing processing cannot be performed on the sheet passing on the sheet conveying path **353**. When stapling or another type of finishing processing is to be performed on a sheet, the sheet receives finishing processing specified by the user in a processing portion **355** via a sheet conveying path **354**, and is discharged to the discharge tray **352**.

[0057] The discharge trays **351** and **352** can each be lifted and dropped. The discharge tray **351** may be dropped down so that a sheet on which finishing processing has been performed in the processing portion **355** can be placed on the discharge tray **351**. When saddle stitch binding is specified, a saddle stitch processing portion **356** performs stapling processing on a central portion of sheets, and then the sheets are folded in half to form a saddle stitch-bound bundle. The saddle stitch-bound bundle is discharged to a saddle stitch binding tray **358** via a sheet conveying path **357**. The saddle stitch binding tray **358** includes a belt conveyor. The saddle stitch-bound bundle placed on the saddle stitch binding tray **358** is conveyed leftward by the belt conveyor.

[0058] The CPU **238** of the inspection device **109** inspects an image of a sheet conveyed thereto in accordance with an inspection item set in advance. The inspection of the image of the sheet is accomplished by a comparison between correct image data stored in the memory **239** in advance and read image data of the image of the conveyed sheet. Examples of the method of comparing the images include a method in which pixel values are compared at each position in the images, a method in which the position of an object is compared by edge detection, and a character data extraction method using optical character recognition (OCR). Examples of the inspection item include a shift in printing position, a hue of the image, a density of the image, streaks and faint printing, stains, and blank spots. Data of those is calculated from the image data read by the inspection device **109**, to be used in determination of the image.

(Inspection Process)

[0059] In the first embodiment, when the same image defect is detected in succession in images on a plurality of sheets (a plurality of recording media) (hereinafter referred to as "continuous NG: No Good") by the inspection device **109**, whether continuous NG has reached a predetermined number of times is determined. When image defects are detected in succession a predetermined number of times, image forming (output) by the printer **107** is stopped. "Continuous NG" refers to successive occurrences of the same type of streak, dot, blank spot, or stain in the same place in read images that are read by the image reading sensors **331** and **332**. Here, continuous NG in a case in which the same image defect in the form of a vertical line is detected in succession is described as an example. When continuous NG reaches a predetermined number of times, whether a factor that is a cause of continuous NG is the printer **107** or the image reading sensors **331** and **332** of the inspection device **109** is determined.

[0060] FIG. **6** is a flow chart of an inspection process. The CPU **238** of the inspection device **109** determines whether the number of times that inspection results are determined to be NG in succession has reached a predetermined number of times in the inspection process. The predetermined number of times may be specified by the user. The CPU **238** functions as a determination unit configured to determine the predetermined number of times based on user's instruction information. When the number of times that inspection results are determined to be NG in succession reaches the predetermined number, the CPU **238** determines whether the successive inspection results determined to be NG are due to the same image defect. When the successive inspection results determined to be NG are due to the same image defect, the CPU **238** executes a factor identification process in which whether a cause of the image defect (a defect factor) is the printer **107** or the image reading sensor **331** is determined. The factor identification process is described later with reference to FIG. **7**.

[0061] The inspection process executed by the CPU **238** as a controller is described below with reference to FIG. **6**. The CPU **238** executes the inspection process in accordance with a program stored in the memory **239**. A sheet on which an image has been formed by the printer **107** is conveyed from the printer **107** to the inspection device **109** via the inserter **108**. The CPU **238** conveys the sheet on which the image is formed to the sheet conveying path **333** (Step S**8101**). The inspection device **109** is provided with the image reading sensor **331**, which is configured to read the top side (first side) of a sheet and is located on an upper side (one side) of the sheet conveying path **333**. The inspection device **109** is provided with the image reading sensor **332**, which is configured to read the bottom side (second side) of a sheet and is located on a lower side (the other side opposite from the one side across the sheet conveying path **333**) of the sheet conveying path **333**. The CPU **238** executes inspection operation in which the top side (first side) of the sheet is inspected by the image reading sensor **331** in time with the conveyance of the sheet (Step S**8102**).

[0062] The CPU **238** executes the inspection process based on image data of the image read by the image reading sensor **331**, and determines whether the inspection result of the inspection process is “OK” or “NG” (Step S**8103**). When the inspection result is OK (“OK” in Step S**8103**), the CPU **238** conveys the sheet to the high capacity stacker **110** so that the sheet is discharged to a discharge tray (the sheet stacking tray **341**, the discharge tray **351**, the discharge tray **352**, or the saddle stitch binding tray **358**) (Step S**8104**). The CPU **238** advances the processing to Step S**8105**.

[0063] When the inspection result is NG (“NG” in Step S**8103**), the CPU **238** increments a value of a counter built in the CPU **238** by 1 (Step S**8106**). The CPU **238** conveys the sheet to the high capacity stacker **110** so that the sheet is discharged to the escape tray **346** of the high capacity stacker **110** (Step S**8107**). The CPU **238** determines whether NG has occurred continuously (Step S**8108**). When determining that NG has not occurred continuously (“NO” in Step S**8108**), the CPU **238** sets the value of the counter to 1 (Step S**8109**), and advances the processing to Step S**8105**. When determining that NG has occurred continuously (“YES” in Step S**8108**), the CPU **238** determines whether the number of times that inspection results are determined to be NG in succession (hereinafter referred to as “the number of continuous NGs”) has reached a predetermined number of times (Step S**8110**). When determining that the number of continuous NGs has not reached the predetermined number of times (“NO” in Step S**8110**), the CPU **238** advances the processing to Step S**8105**.

[0064] A setting screen **141** on which the number of continuous NGs is set is described. FIG. **12** is a schematic diagram of the setting screen **141** to be displayed on the display portion **241**. When a command issued by the user to display the setting screen **141** is received from the operation portion **242**, the CPU **238** displays the setting screen **141** illustrated in FIG. **12** on the display portion **241**. The setting screen **141** displays radio buttons for setting whether printing and inspection are to be cancelled when the number of continuous NGs reaches the predetermined number of times. The setting screen **141** displays a field in which the predetermined number of times can be input. Selection of the radio button for cancellation enables the user to input to this field. When the user chooses to cancel printing and inspection, the CPU **238** receives a number input to the field. The CPU **238** receives user's instruction information about the predetermined number of times input to the field described above, and sets the numerical value input as the predetermined number of times as well. When the user chooses not to cancel printing and inspection on the setting screen **141**, the CPU **238** do not cancel inspection and printing no matter how many times NG occurs in succession.

[0065] When determining that the number of continuous NGs has reached the predetermined number of times ("YES" in Step S**8110**), the CPU **238** determines whether the inspection results that are NG are due to the same image defect (the same defective phenomenon) (Step S**8111**). Specifically, whether inspection results that are NG are due to the same image defect is determined based on NG determination classification (image quality information) of inspection and position information of an NG occurrence point. When continuous NG varies in NG determination classification or in position information of an NG occurrence point, it is determined that the inspection results that are NG are not due to the same image defect ("NO" in Step S**8111**). The CPU **238** resets the value of the counter to 0 (Step S**8112**).

[0066] The CPU **238** advances the processing to Step S**8105**. The CPU **238** determines whether there is an inspection waiting sheet (Step S**8105**). When there is an inspection waiting sheet ("YES" in Step S**8105**), the CPU **238** returns the processing to Step S**8102** to continue the inspection operation. When there is no inspection waiting sheet ("NO" in Step S**8105**), the CPU **238** ends the inspection process.

[0067] When continuous NG is the same in NG determination classification and in position information of an NG occurrence point, it is determined that the inspection results that are NG are due to the same image defect ("YES" in Step S**8111**). The CPU **238** transmits an image formation stop notification to the CPU **222** of the printer **107** (Step S**8113**). In Step S**8113**, the CPU **238** displays a cancellation screen **142** illustrated in FIG. **13** on the display portion **241**. The cancellation screen **142** displays a message for notifying that inspection has been cancelled halfway. When the user presses an OK button on the cancellation screen **142**, the CPU **238** shifts the screen displayed on the display portion **241** from the cancellation screen **142** to a screen for displaying an inspection result. The CPU **238** executes the factor identification process in which whether a cause of the image defect (defect factor) is the printer **107** or the image reading sensor **331** is determined (Step S**8114**).

(Factor Identification Process)

[0068] A sub-routine of the factor identification process executed in Step S**8114** of FIG. **6** is described below with reference to FIG. **7**. FIG. **7** is a flow chart of the factor identification process. In the factor identification process, the CPU **238** determines whether the cause of the image defect is the printer **107** or the image reading sensors **331** and **332**. First, the CPU **238** determines whether the NG determination classification of continuous NG is a vertical line (Step S**9001**). When the NG determination classification of continuous NG is not the vertical line ("NO" in Step S**9001**), the CPU **238** determines that the cause of the image defect is the printer **107**, and notifies the defect to the CPU **222** of the printer **107** (Step S**9002**). The CPU **238** ends the factor identification process.

[0069] When the NG determination classification of continuous NG is the vertical line ("YES" in Step S**9001**), the CPU **238** instructs the printer **107** to form a sample image (Step S**9003**). The CPU **238** instructs the inserter **108** to execute a side shift to shift sideways a sheet on which the sample image has been formed, with the use of the shift mechanism of the inserter **108** (Step S**9004**). The shift mechanism of the inserter **108** executes a side shift so that the sheet is shifted by 5 mm in a direction orthogonal to the sheet conveyance direction. In the first embodiment, the sheet is shifted rightward by 5 mm, but the shift mechanism is not limited thereto. The sheet may be shifted rightward by 6 mm or 4 mm, or may be shifted leftward by 5 mm. The CPU **238** executes inspection operation for the sheet with the sample image formed thereon (Step S**9005**).

[0070] The CPU **238** determines whether position information of the vertical line in the sample image is the same as the position information of the NG occurrence point of continuous NG, based on the result of the inspection process (Step S**9006**). When the position information of the vertical line in the sample image is the same as the position information of the NG occurrence point of continuous NG ("YES" in Step S**9006**), the CPU **238** determines that the cause of the vertical line is the printer **107**, and notifies the defect to the CPU **222** of the printer **107** (Step S**9007**). The CPU **238** ends the factor identification process.

[0071] When the position information of the vertical line in the sample image differs from the position information of the NG occurrence point of continuous NG ("NO" in Step S**9006**), the CPU **238** determines that the cause of the vertical line is the image reading sensor **331** (Step S**9008**). The CPU **238** ends the factor identification process.

[0072] In the first embodiment, the factor identification process is executed when an image formed on one side of a sheet is inspected by the inspection device **109** and the number of times that inspection results are determined to be NG in succession reaches the predetermined number of times. When an image defect that is a vertical line is detected at the same position in succession the predetermined number of times, the inspection results of continuous NG and the inspection result of the sample image are compared. Whether the cause of NG is the printer **107** or the image reading sensor **331** is determined based on the result of the comparison. According to the first embodiment, when the same image defect is detected in succession in images on recording media, whether the cause of NG is the printer **107** or the image reading sensor **331** can be identified. According to the first embodiment, continuation of output of printed products that are determined to be errors as a result of inspection can be inhibited.

Second Embodiment

[0073] Description is made of a second embodiment. In the second embodiment, structure that is the same as the one in the first embodiment is denoted by the same reference symbols and numerals, and description thereof is omitted. The printing system **100** and an inspection process in the second embodiment are the same as those in the first embodiment, and description thereof is therefore omitted. A factor identification process in the second embodiment differs from the one in the first embodiment. The factor identification process in the second embodiment is described below.

(Factor Identification Process)

[0074] A sub-routine of the factor identification process of the second embodiment executed in Step S**8114** of FIG. **6** is described below with reference to FIG. **8**. FIG. **8** is a flow chart of the factor identification process of the second embodiment. The CPU **238** determines whether the NG determination classification of continuous NG is the vertical line (Step S**4001**). When the NG determination classification of continuous NG is not the vertical line ("NO" in Step S**4001**), the CPU **238** determines that the cause of continuous NG is the printer **107**, and notifies the defect to the CPU **222** of the printer **107** (Step S**4002**).

[0075] When the NG determination classification of continuous NG is the vertical line ("YES" in Step S**4001**), the CPU **238** conveys a sheet on which an image has been formed and which remains on the sheet conveying path **312**, **314**, or **315** of the printer **107** (Step S**4003**). The CPU **238** instructs the inserter **108** to execute a side shift to shift sideways the sheet with the image formed thereon by the shift mechanism of the inserter **108** (Step S**4004**). The side shift is not limited to the shift mechanism of the inserter **108**. The side shift may be executed on a sheet conveying path by sliding the conveyance roller of the printer **107** in a side direction. The CPU **238** executes inspection operation for the sheet with the image formed thereon (Step S**4005**).

[0076] The CPU **238** determines whether position information of a vertical line in the image of the remaining sheet is the same as the position information of the NG occurrence point of continuous

NG, based on the result of the inspection process (Step S**4006**). When the position information of the vertical line in the image of the remaining sheet differs from the position information of the NG occurrence point of continuous NG ("NO" in Step S**4006**), the CPU **238** determines that the cause of the vertical line is the image reading sensor **331** (Step S**4007**). The inspection device **109** displays alert to the user by displaying recovery (automatic restoration) procedures of the image reading sensor **331**. For example, the inspection device **109** displays a prompt for the user to clean the image reading sensor **331** on the display portion **241** (Step S**4008**).

[0077] The CPU **238** checks whether the user has cleaned the image reading sensor **331** (Step S**4009**). When it is confirmed that the image reading sensor **331** has been cleaned, the CPU **238** feeds a sheet from the inserter **108** and executes inspection operation (Step S**4010**). The CPU **238** determines whether the inspection result is OK or NG (Step S**4011**). When the same vertical line occurs and the inspection result is NG ("NG" in Step S**4011**), it is highly probable that a failure has occurred in the image reading sensor **331**, and the CPU **238** accordingly contacts a support desk (Step S**4012**). When the inspection result is OK ("OK" in Step S**4011**), on the other hand, the CPU **238** restarts image formation by the printer **107** (Step S**4013**).

[0078] When the position information of the vertical line in the image of the remaining sheet is the same as the position information of the NG occurrence point of continuous NG ("YES" in Step S**4006**), on the other hand, the CPU **238** determines that the cause of the vertical line is the printer **107** and notifies the printer **107** of the defect (Step S**4002**). The CPU **238** displays recovery procedures on the printer **107** (Step S**4014**). The CPU **238** checks whether the user has finished the recovery procedures of the printer **107** (Step S**4015**). When the end of the recovery procedures has been confirmed, the CPU **238** instructs the printer **107** to form a test image (Step S**4016**). The CPU **238** executes inspection operation for the test image (Step S**4017**).

[0079] The CPU **238** determines whether the inspection result is OK or NG (Step S**4018**). When the same vertical line occurs and the inspection result is NG ("NG" in Step S**4018**), the CPU **238** contacts the support desk (Step S**4012**). When the inspection result is OK ("OK" in Step S**4018**), on the other hand, the CPU **238** restarts image formation by the printer **107** (Step S**4013**).

[0080] According to the second embodiment, when the same image defect is detected in images on recording media in succession, whether the cause of NG is the printer **107** or the image reading sensor **331** can be identified. Diagnosis of recovery (automatic restoration) of the identified printer **107** or image reading sensor **331** is executed and, when the printer **107** or the image reading sensor **311** is restored, image formation by the printer **107** is restarted. When restoration fails, the support desk can be contacted. According to the second embodiment, continuation of output of printed products that are determined to be errors as a result of inspection can be inhibited.

Third Embodiment

[0081] Description is made of a third embodiment. In the third embodiment, structure that is the same as the one in the first embodiment is denoted by the same reference symbols and numerals, and description thereof is omitted. The printing system **100** and an inspection process in the third embodiment are the same as those in the first embodiment, and description thereof is therefore omitted. A factor identification process in the third embodiment differs from those in the first and second embodiments. The factor identification process in the third embodiment is described below. In the third embodiment, when the image reading sensor (first sensor) **331** detects an image defect that is a vertical line at the same position on the top side (first side) of a sheet in succession, the sheet is conveyed to the sheet reversing path **316** in the printer **107** to be switched back. The top side (first side) and bottom side (second side) of the sheet are reversed, and the image reading sensor (second sensor) **332** reads an image of the top side (first side) of the sheet. Inspection results of continuous NG and an inspection result of the image reading sensor **332** are compared. Whether the cause of NG is the printer **107** or the image reading sensor **331** is determined based on the result of the comparison.

(Factor Identification Process)

[0082] A sub-routine of the factor identification process in the third embodiment to be executed in Step S**8114** of FIG. **6** is described below with reference to FIG. **9**. FIG. **9** is a flow chart of the factor identification process in the third embodiment. The CPU **238** determines whether the NG determination classification of continuous NG is the vertical line (Step S**5001**). When the NG determination classification of continuous NG is not the vertical line ("NO" in Step S**5001**), the CPU **238** determines that the cause of the image defect is the printer **107**, and notifies the defect to the CPU **222** of the printer **107** (Step S**5002**).
[0083] When the NG determination classification of continuous NG is the vertical line ("YES" in Step S**5001**), the CPU **238** conveys a sheet on which an image has been formed and which remains on the sheet conveying path **312**, **314**, or **315** of the printer **107** to the sheet reversing path **316** (Step S**5003**). The sheet is switched back by the sheet reversing path **316** to reverse the front and back of the sheet. The CPU **238** conveys the sheet with the front and the back reversed to the inspection device **109** (Step S**5004**). The CPU **238** uses the image reading sensor (second sensor) **332** placed on the side below the sheet conveying path **333** to execute inspection operation for the sheet whose top side (first side) is facing downward (Step S**5005**).
[0084] The CPU **238** determines based on the result of the inspection process whether position information of a vertical line in the image of the switched back sheet is the same as the position information of the NG occurrence point of continuous NG (Step S**5006**). When the vertical line is detected at different positions in the inspection result of the image reading sensor (first sensor) **331** and the inspection result of the image reading sensor (second sensor) **332**, it can be determined that the cause of continuous NG is the image reading sensor (first sensor) **331**. When the vertical line is detected at the same position in the inspection result of the image reading sensor (first sensor) **331** and the inspection result of the image reading sensor (second sensor) **332**, it can be determined that the cause of continuous NG is the printer **107**. The CPU **238** accordingly executes processing as follows.
[0085] When position information of the vertical line in the image of the switched back sheet differs from the position information of the NG occurrence point of continuous NG ("NO" in Step S**5006**), the CPU **238** determines that the cause of the vertical line is the image reading sensor **331** (Step S**5007**). The inspection device **109** displays alert to the user by displaying recovery (automatic restoration) procedures of the image reading sensor **331**. For example, the inspection device **109** displays a prompt for the user to clean the image reading sensor **331** on the display portion **241** (Step S**5008**).
[0086] The CPU **238** checks whether the user has cleaned the image reading sensor **331** (Step S**5009**). When it is confirmed that the image reading sensor **331** has been cleaned, the CPU **238** feeds a sheet from the inserter **108** and executes inspection operation (Step S**5010**). The CPU **238** determines whether the inspection result is OK or NG (Step S**5011**). When the same vertical line occurs and the inspection result is NG ("NG" in Step S**5011**), it is highly probable that a failure has occurred in the image reading sensor **331**, and the CPU **238** accordingly contacts the support desk (Step S**5012**). When the inspection result is OK ("OK" in Step S**5011**), on the other hand, the CPU **238** restarts image formation by the printer **107** (Step S**5013**).
[0087] When the position information of the vertical line in the image of the switched back sheet is the same as the position information of the NG occurrence point of continuous NG ("YES" in Step S**5006**), on the other hand, the CPU **238** determines that the cause of the vertical line is the printer **107** and notifies the printer **107** of the defect (Step S**5002**). The CPU **238** displays recovery procedures on the printer **107** (Step S**5014**). The CPU **238** checks whether the user has finished the recovery procedures of the printer **107** (Step S**5015**). When it is confirmed that the end of the recovery procedures has been confirmed, the CPU **238** instructs the printer **107** to form a test image (Step S**5016**). The CPU **238** executes inspection operation for the test image (Step S**5017**).
[0088] The CPU **238** determines whether the inspection result is OK or NG (Step S**5018**). When the same vertical line occurs and the inspection result is NG ("NG" in Step S**5018**), the CPU **238**

contacts the support desk (Step S**5012**). When the inspection result is OK ("OK" in Step S**5018**), on the other hand, the CPU **238** restarts image formation by the printer **107** (Step S**5013**).

[0089] According to the third embodiment, when the same image defect is detected in images on recording media in succession, whether the cause of NG is the printer **107** or the image reading sensor **331** can be identified. Diagnosis of recovery (automatic restoration) of the identified printer **107** or image reading sensor **331** is executed and, when the printer **107** or the image reading sensor **311** is restored, image formation by the printer **107** is restarted. When restoration fails, the support desk can be contacted. According to the third embodiment, continuation of output of printed products that are determined to be errors as a result of inspection can be inhibited.

Fourth Embodiment

[0090] Description is made of a fourth embodiment. In the fourth embodiment, structure that is the same as the one in the first embodiment is denoted by the same reference symbols and numerals, and description thereof is omitted. The printing system **100** and an inspection process in the fourth embodiment are the same as those in the first embodiment, and description thereof is therefore omitted. In the fourth embodiment, the printer **107** forms an image on each side of a sheet in a double-sided printing mode. A factor identification process in the fourth embodiment is executed when images formed on both sides of a sheet are to be inspected. The factor identification process in the fourth embodiment differs from those in the first to third embodiments. The factor identification process in the fourth embodiment is described below.

(Factor Identification Process)

[0091] A sub-routine of the factor identification process in the fourth embodiment to be executed in Step S**8114** of FIG. **6** is described below with reference to FIG. **10**. FIG. **10** is a flow chart of the factor identification process in the fourth embodiment. The CPU **238** determines whether the NG determination classification of continuous NG is the vertical line in any one of the NG determination classification of the image reading sensor (first sensor) **331** and the NG determination classification of the image reading sensor (second sensor) **332** (Step S**6001**). When none of the NG determination classification of the image reading sensor (first sensor) **331** and the NG determination classification of the image reading sensor (second sensor) **332** is the vertical line ("NO" in Step S**6001**), the CPU **238** determines that the cause of continuous NG is the printer **107**, and notifies the defect to the CPU **222** of the printer **107** (Step S**6002**).

[0092] When at least one of the NG determination classification of the image reading sensor **331** or the NG determination classification of the image reading sensor **332** is the vertical line ("YES" in Step S**6001**), the CPU **238** determines whether the number of continuous NGs has reached the predetermined number of times with both of the image reading sensors **331** and **332** (Step S**6003**). When determining that the number of continuous NGs has reached the predetermined number of times with only one of the image reading sensors **331** and **332** ("NO" in Step S**6003**), the CPU **238** determines that the one of the image reading sensors **331** and **332** is defective (Step S**6004**). When determining that the number of continuous NGs has reached the predetermined number of times with both of the image reading sensors **331** and **332** ("YES" in Step S**6003**), the CPU **238** determines whether the NG determination classification of the image reading sensor **331** and the NG determination classification of the image reading sensor **332** are both vertical line (Step S**6005**).

[0093] When the NG determination classification of continuous NG differs between the image reading sensors **331** and **332** ("NO" in Step S**6005**), the CPU **238** determines that one image reading sensor and the printer **107** are both defective (Step S**6006**). The image reading sensor determined to be defective is the one for which it is determined that the NG determination classification of continuous NG is the vertical line. A defect in the image read by the image reading sensor for which it is determined that the NG determination classification is not the vertical line is determined to be caused by the printer **107**. The CPU **238** notifies the printer **107** of the defect (Step S**6007**).

[0094] When the NG determination classification is the vertical line for both of the image reading sensors **331** and **332** ("YES" in Step S**6005**), the CPU **238** determines whether the image reading sensors **331** and **332** have the same position information of the NG occurrence point (Step S**6008**). When the image reading sensors **331** and **332** have the same position information of the NG occurrence point ("YES" in Step S**6008**), the CPU **238** determines that the printer **107** is defective, and notifies the printer **107** of the defect (Step S**6009**). When the image reading sensors **331** and **332** differ from each other in position information of the NG occurrence point ("NO" in Step S**6008**), the CPU **238** determines that the image reading sensors **331** and **332** are both defective (Step S**6010**).
[0095] The factor identification process in the fourth embodiment is executed when images formed on both sides of a sheet by the printer **107** with the image forming mode set to the double-sided printing mode are to be inspected. According to the fourth embodiment, when the same image defect is detected in succession in images on recording media, whether the cause of NG is the printer **107**, the image reading sensors **331** and **332**, or both of the printer **107** and the image reading sensors **331** and **332** can be identified. According to the fourth embodiment, continuation of output of printed products that are determined to be errors as a result of inspection can be inhibited.
Fifth Embodiment
[0096] Description is made of a fifth embodiment. In the fifth embodiment, structure that is the same as the one in the first embodiment is denoted by the same reference symbols and numerals, and description thereof is omitted. The printing system **100** and an inspection process in the fifth embodiment are the same as those in the first embodiment, and description thereof is therefore omitted. In the fifth embodiment, improvement operation is executed when the cause of NG is identified as the printer **107**, the image reading sensor **331**, or both of the printer **107** and the image reading sensor **331** in the factor identification processes of the first to fourth embodiments. The improvement operation in the fifth embodiment is described below.
(Improvement Operation)
[0097] FIG. **11** is a flow chart of the improvement operation in the fifth embodiment. The CPU **238** determines whether defects occur in both of the printer **107** and the image reading sensor **331** or **332** (Step S**7001**). When a defect occurs in one of the printer **107** and the image reading sensor **331** or **332** ("NO" in Step S**7001**), the CPU **238** determines whether the defect occurs in the image reading sensor **331** or **332** (Step S**7002**). When the defect does not occur in the image reading sensor **331** or **332** ("NO" in Step S**7002**), the CPU **238** determines that the printer **107** is defective (Step S**7003**). The CPU **238** notifies the printer **107** of the defect (Step S**7004**). The CPU **238** makes the printer **107** execute improvement control for improving a vertical line occurrence factor (Step S**7005**). Specifically, the printer **107** executes cleaning of a wire of a charger and belt-pattern application control in which a toner image is formed in a belt-like shape in order to prevent a toner from slipping along a cleaning member. The CPU **238** ends the improvement operation.
[0098] When a defect occurs in the image reading sensor **331** or **332** ("YES" in Step S**7002**), the CPU **238** determines that the image reading sensor **331** or **332** is defective (Step S**7006**). The CPU **238** executes improvement control for improving the image reading sensor **331** or **332** (Step S**7007**). Specifically, an output trouble of the image reading sensor **331** or **332** is improved through a removal of a foreign object adhering to the image reading sensor **331** or **332** by putting a cleaning member of the image reading sensor **331** or **332** into operation. The CPU **238** ends the improvement operation. When defects occur in both of the printer **107** and the image reading sensor **331** or **332** ("YES" in Step S**7001**), the CPU **238** notifies the printer **107** of the defect (Step S**7008**). The CPU **238** makes the printer **107** execute improvement control for improving a vertical line occurrence factor (Step S**7009**). At the same time, the CPU **238** executes improvement control of the image reading sensor **331** or **332** (Step S**7010**). The CPU **238** ends the improvement operation.

[0099] In the fifth embodiment, the cleaning of the wire of the charger and belt-pattern application control in which a toner image is formed in a belt-like shape in order to prevent a toner from slipping along the cleaning member are given as examples of the improvement control executed by the printer **107** to improve a vertical line occurrence factor. However, the improvement control executed by the printer **107** to improve a vertical line occurrence factor is not limited to those, and may be another form of control for preventing occurrence of a vertical line. In the fifth embodiment, control for putting the cleaning member of the image reading sensor **331** or **332** into operation is described as an example of the improvement control of the image reading sensor **331** or **332**. However, the improvement control of the image reading sensor **331** or **332** is not limited thereto, and may be another form of control for improving an output trouble of the image reading sensor **331** or **332**.
[0100] As described above, the image forming apparatus **101** can be brought back from an image formation ceasing state to a normal state by executing the improvement control after identifying which of the printer **107** and the image reading sensor **331** or **332** is the cause of NG. In the fifth embodiment, after the improvement control is executed, a print job may be resumed from an image determined to be NG in the inspection process, and the inspection process may then be performed. When a defect (streak or dot) of the same type as the one in an inspection result prior to the execution of the improvement control appears at the same position in an inspection result of the image immediately after the resumption, alert may be displayed on the display portion **225** of the printer **107** at the same time as the image forming operation of the printer **107** is stopped. The inspection device **109** may be controlled so as to hold off the execution of the inspection process by the inspection device **109** until a service person performs a checkup.
[0101] According to the fifth embodiment, when inspection results indicate NG in succession, a factor of NG is identified and diagnosis on whether automatic restoration is possible can be performed. According to the fifth embodiment, when images formed in succession have the same image defect, whether the cause of the image defect is the printer **107** or one of the image reading sensors **331** and **332** can be identified. According to the fifth embodiment, continuation of output of printed products determined to be errors as a result of inspection can be inhibited.
[0102] The inspection device **109** in the fifth embodiment includes the CPU **238** and the display portion **241**, which function as an information processing apparatus. However, the inspection device **109** may be connected, via a network, to an information processing apparatus that is a device separate from the inspection device **109**. For example, a configuration may be employed in which the CPU **208** and display **212** of the external controller **102** function as an information processing apparatus. An alternative configuration may be employed in which the CPU **201** and display **205** of the PC **103** function as an information processing apparatus.
[0103] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
[0104] This application claims the benefit of Japanese Patent Application No. 2020-028538, filed Feb. 21, 2020, which is hereby incorporated by reference herein in its entirety.

Claims

1. An information processing apparatus which is communicable in communication to and from an image forming apparatus configured to form an image on a sheet, and in communication to and from a reader configured to read the image on the sheet, the reader being connected to the image forming apparatus, the information processing apparatus comprising: a display configured to display an inspection result; and a controller configured to: obtain read image data output from the reader; determine an error of the read image data based on the read image data and reference data registered in advance; cancel image forming operation of forming an image by the image forming apparatus in a case in which the error is determined in succession a predetermined number of times; cancel reading operation of reading an image by the reader in the case in which the error is determined in succession the predetermined number of times; notify, on the display, the cancellation of the image forming operation and the reading operation in the case in which the error is determined in succession the predetermined number of times; and receive a user's instruction information about the predetermined number of times.